

Exercise 1. Let $C^1[0, 1]$ denote the space of all continuously differentiable functions on $[0, 1]$. For $f, g \in C^1[0, 1]$ set

$$d_1(f, g) = \max_{t \in [0, 1]} |f(t) - g(t)|,$$

and

$$d_2(f, g) = \max_{t \in [0, 1]} |f(t) - g(t)| + \max_{t \in [0, 1]} |f'(t) - g'(t)|.$$

Show that:

- a) d_1, d_2 are both metrics on $C^1[0, 1]$,
- b) $(C^1[0, 1], d_2)$ is complete while $(C^1[0, 1], d_1)$ is not.

Exercise 1. Let $\{(X_n, d_n) : n \geq 1\}$ be a countable family of metric spaces. Show that the function

$$d : \prod_{n \geq 1} X_n \times \prod_{n \geq 1} X_n \rightarrow R_+$$

defined by

$$((x_n), (y_n)) = \sum_{n \geq 1} 2^{-n} \frac{d_n(x_n, y_n)}{1 + d_n(x_n, y_n)}$$

defines a metric on $\prod_{n \geq 1} X_n$. Also show that if each (X_n, d_n) is complete then $(\prod_{n \geq 1} X_n, d)$ is complete.

Exercise 3. Consider the set

$$Y = \{f \in C([0, 1]) : f \text{ is differentiable at } \frac{1}{2}\}.$$

Show that Y is of the first category in $C([0, 1])$.

In the exercises below, for $1 \leq p \leq \infty$ the space l^p consists of sequences (x_n) of complex numbers with

$$\sum_n |x_n|^p < \infty, \quad \text{if } p < \infty,$$

and

$$\sup_n |x_n| < \infty, \quad \text{if } p = \infty.$$

It is easy to see that (lecture!)

$$d_p(x, y) = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{\frac{1}{p}}, \quad \text{if } 1 \leq p < \infty \text{ and } d_{\infty}(x, y) = \sup_{n \geq 1} |x_n - y_n|,$$

is a metric on l^p . We have seen in the lecture that each of these spaces is complete.

Exercise 4. Prove that for $1 \leq p < \infty$, l^p is separable, but l^∞ is not separable.

Exercise 5. Show that in the space l^1 (with the usual metric) a set with interior cannot be compact.

Exercise 6. Show that a set $\mathcal{F} \subset l^p$, $1 \leq p < \infty$ is relatively compact if and only if for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that for every $x = (x_n) \in \mathcal{F}$, and every $n > n_\varepsilon$ we have

$$\sum_{k \geq n} |x_k|^p < \varepsilon.$$

Exercise 7. Consider the linear map $V : C([0, 1]) \rightarrow C([0, 1])$ defined by

$$Vf(x) = \int_0^x f(t) dt.$$

Show that V maps any bounded set in $C([0, 1])$ onto a relatively compact set.